

UI/UX for Planetarium & GNSS

Module 1: User Research & User Flow

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1. Objectives

- Identify the distinct user groups who will interact with the Planetarium & Observatory Management System.
- Define what each user group needs from the platform and the pain they currently face when trying to access similar services elsewhere.
- Map a basic flow showing how each user type is expected to move through the system, from entry point to task completion.
- Use these findings to guide the wireframing, information architecture, and screen-level decisions in the modules that follow.

User Type	Description	Goals	Pain Points	Key Pages / Features
Students	School/college groups and individual learners visiting for structured educational content.	Book field trips easily; find programs matched to their grade/age level; access supporting material before or after a visit.	Program listings not filtered by age group; unclear group booking process; no confirmation of session inclusions.	Programs page (audience-tagged), group booking form, schedule
Visitors (General Public)	Casual and tourist visitors, families, and first-time guests exploring what the planetarium offers.	Quickly check show times, book tickets with minimal steps, find practical visit info (hours, location, pricing).	Booking flows with too many fields; unclear booking confirmation; navigation cluttered with content not meant for them.	Homepage quick links, show schedule, ticket booking, visit-planning info
Researchers	Academic and scientific users engaging with observatory data, telescope access, or published research.	Reserve telescope sessions, browse research publications, contact relevant staff for collaboration.	Research content buried under public-facing marketing pages; no dedicated request form for telescope access.	Research portal, publication archive, telescope session request form
Faculty	Educators arranging visits for their students or referencing planetarium content for coursework.	Compare programs suited to class size and curriculum; secure group pricing; get confirmation early enough to plan around it.	No clear distinction between individual and group booking; limited program detail for planning purposes.	Programs page, group/field-trip booking flow, downloadable program info

User Type	Description	Goals	Pain Points	Key Pages / Features
Staff	Front-desk and event-day operations staff supporting day-to-day activities.	View the day's bookings and schedule at a glance; make quick manual adjustments when needed (walk-ins, cancellations).	Admin tools built for full administrators, not simple day-to-day lookups.	Simplified operations view within the dashboard, booking lookup
Administrators	Internal system managers are responsible for content, scheduling, and oversight across the platform.	Manage bookings, update programs/schedules, monitor activity without needing developer support.	No centralized view of bookings across programs; manual updates required for simple content changes.	Admin dashboard, program/schedule management, content editor

2. Methodology

User groups, needs, and flows for this module were identified by combining the project brief and stakeholder requirements with findings from the competitor analysis previously conducted for this project, which examined the Shanghai Astronomy Museum and Adler Planetarium websites and is documented separately. That report is referenced here only where its findings directly informed a decision below — for example, Adler's audience-tagged programs and isolated, distraction-free checkout flow were both considered when defining the user needs and flows that follow.

3. Target Users

The table below summarizes the primary user groups for the platform, their goals, the pain points they typically encounter, and the parts of the system most relevant to them.

4. User Journey Flows

The flows below outline, at a high level, how each core user type is expected to move through the system — from arriving on the site to complete their primary task.

4.1 Visitor Flow (Booking a Show)

Homepage



Show Schedule



Select Session



Enter Visitor Details



Booking Confirmation

4.2 Student Flow (via Faculty / Group Booking)

Homepage / Programs Page



Filter by Audience (School Groups)



View Program Details & Group Pricing



Submit Group Booking Request



Confirmation & Follow-up

4.3 Researcher Flow (Telescope Session & Publications)

Homepage



Research Portal



Browse Publications / Research Areas



Telescope Session Request



Confirmation & Access Details

4.4 Administrator Flow (Managing the System)

Login



Admin Dashboard



Review Bookings & Schedules



Approve / Edit / Update Content



Changes Reflected on Live Site

5. Sitemap

A high-level sitemap consolidating the sections and pages referenced across the user flows above.

Home



— About



— Experiences



— Dome Shows

— Night Sky Tours

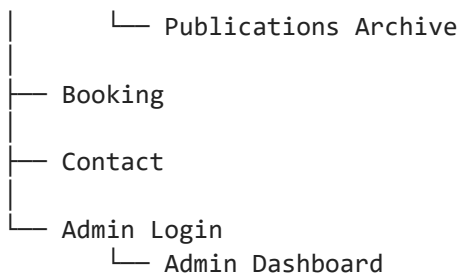
— School Programs



— Observatory



— Research



6. Key UX Insights

- Keep the booking process as short as possible — every extra field or screen is another chance for a visitor to drop off before confirming.
- Apply one consistent CTA hierarchy across the site: a single visual style for primary actions like booking, and a distinct style for secondary actions like donations or contact, so users always know what's expected of them.
- Separate the visitor journey from the researcher and administrator journeys early, rather than routing them through the same navigation — their goals rarely overlap.
- Tag content by audience (students, researchers, general public) so each group can filter to what's relevant instead of scanning an undifferentiated list.
- Maintain one consistent navigation structure across the site rather than changing layout patterns section to section, so the platform feels like a single system rather than several linked pages.
- Minimize the number of steps required to complete key user tasks, particularly booking.
- Establish a clear information hierarchy on every page — one primary heading, one primary action, and supporting content kept visually secondary.

7. Design Implications

This research directly shapes the wireframes and UI currently in progress. The homepage now under design in Figma follows the audience-clarity and hierarchy principles above — visitors are guided toward a single primary action, while secondary paths such as About, Experiences, and Observatory stay visually distinct without competing for attention. The booking-flow structure identified for visitors and students will inform the layout of the upcoming Experiences and booking screens, and the distraction-free approach noted in the competitor research will be applied wherever a user is mid-booking. These decisions will carry through consistently into the wireframes and high-fidelity screens produced in the next stage of this module.

8. Next Steps

- Continue high-fidelity UI design of the Planetarium homepage.
- Develop low-fidelity wireframes for the booking page and observatory page.
- Design the booking flow in detail.
- Begin wireframes for the admin dashboard.
- Validate all screens against the user flows and sitemap in this document before moving into prototyping.